

JEE MAIN REVISION MASTER: PROPERTIES OF MATTER SYLLABUS REDUCED

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 301 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Properties Of Matter - Core Application	301 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Properties Of Matter - Core Application (Very High Yield — 301 PYQs)

Core Principle: Key problem pattern in Properties Of Matter.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "A liquid of density 600kg/m^3 flowing steadily in a tube of varying cross-section. The cross-section at a point A is 1.0cm^2 and t..." [Ans: D]
- [2026 Morning Shift] Q#1 (MCQ): "A lift of mass 1600 kg is supported by thick iron wire. If the maximum stress which the wire can withstand is $4 \times 10^8\text{N/m}^2$ and i..." [Ans: A]
- [2026 Morning Shift] Q#1 (MCQ): "Two wires as shown in the figure below, made of steel and have breaking stress of $12 \times 10^8\text{N/m}^2$. Area of cross-section of upper..." [Ans: B]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

A liquid of density 600kg/m^3 flowing steadily in a tube of varying cross-section. The cross-section at a point A is 1.0cm^2 and that at B is 20mm^2 . Both the points A and B are in same horizontal plane, the speed of the liquid at A is 10cm/s . The difference in pressures at A and B points is ____ Pa.

- (A) 18
- (B) 144
- (C) 36
- (D) 72 [Correct]

Correct Answer: D

Solution Analysis: The density of liquid is $\rho=600\text{kg/m}^3$. Area of cross-section at point A is $A_A=1.0\text{cm}^2=1.0 \times 10^{-4}\text{m}^2$. Area of cross-section at point B is $A_B=20.0\text{mm}^2=20.0 \times 10^{-6}\text{m}^2=0.2 \times 10^{-4}\text{m}^2$. The velocity at A is $v_A=10\text{cm/s}=0.1\text{m/s}$. For an incompressible liquid in steady flow, the mass flow rate must remain constant. Using equation of continuity, $A_A v_A = A_B v_B$ Substituting the values : $(1.0 \times 10^{-4}) \times (0.1) = (0.2 \times 10^{-4}) \times v_B \Rightarrow v_B = (1.0 \times 0.1 / 0.2) = (0.1 / 0.2) = 0.5\text{m/s}$ According to